

An Artist-Friendly Method for Procedural Skin Generation and Visualisation in Houdini

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Figure 1: Samples of skin microstructures created with our method and rendered using V-Ray in Houdini.

ABSTRACT

Realistic-looking digital humans in visual effects are a crucial aspect of creating a believable experience for the viewer. In order to create a convincing result, a lot of effort is put into the details. In this talk, we look specifically into the details of the human skin and how we implemented a fully procedural approach to generate and visualise skin textures. The tool is integrated into our VFX-pipeline as a SOP (surface operator) in Houdini and provides an artist-friendly interface with many ways to modify and tweak the output appearance. Additionally, we provide a way to utilise the underlying animation to produce pores and wrinkles dynamically driven by stretch and compression in the current frame.

CCS CONCEPTS

• Computing methodologies → Computer graphics.

KEYWORDS

skin microstructure, procedural generation, digital humans

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1 INTRODUCTION

When scanning digital characters, it is only possible to capture finer geometry details down to a certain level. This leaves out very small pores and microstructures that greatly contribute to the realism

of the digital character. Thus, there's a need to be able to recreate the micro-level details of the skin surface using other methods than scanning. We present an approach to procedurally generate skin textures and micro-details for our production workflow using Houdini.

2 METHOD

The main parts of our method consist of pore and wrinkle generation. On top of this, we provide a visualisation mode in the Houdini viewport, and options to use the underlying tension in an animated sequence to drive the pore and wrinkle generation.

2.1 Pore Generation

We start the skin generation by scattering the pore positions on the input mesh using a blue noise point distribution with recursive Wang tiles [Kopf et al. 2006]. As input, we provide a mesh and a density map that encodes the ratio between the area in world space and texture space for each face in the input mesh. We let this act as a multiplier in the point distribution which gives us more pores placed closer together in areas of high density, and fewer pores with larger separation in areas of low density.

We further determine the size and the orientation of the pores, where we model each pore as an ellipse represented by its center point, two axes, and an orientation in which the pore is pointing. The position of the pores comes directly from the point distribution. The size of the pores is partially driven using the density map as described above but can be tweaked further by user input or driven dynamically using the underlying mesh tension.

When using the tension of the mesh, we retrieve the stretch and compression using the method of Nagano et al. [2015] which we have tweaked to adapt to our production workflow in Houdini [Haapaoja and Klintberg 2022]. If there is no tension in a sampled point on the mesh, we model the pore as a circle with equal axes. If we have tension, we stretch the pore in the direction of the stretch and compress it in the direction of the compression. We always set the orientation of the pore to be in the direction of stretch.

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2.2 Wrinkle Generation

In the second part of the skin generation algorithm, we generate wrinkles. Similar to Weiss et al. [2023], we want the wrinkles to start and end in the previously determined pore positions and we use the pores as a starting point to generate our wrinkle connections. This means that we always generate pore positions, even in the case of having a pore size of 0, to be able to generate the wrinkle connections between them.

We model each wrinkle with a start position and an end position in 2D space and introduce wrinkle connections where we store indices of pores connected to each other. To determine which pores to connect, we add all 2D pore positions to a k-d tree. We iterate over all pore positions, query the closest pores, and connect them through a wrinkle. We let the user define how many wrinkle connections to generate per pore, but the number can be fewer if the closest pores are outside of our search radius.

When utilising the dynamic properties of the mesh, and a pore is exposed to tension, we offset the search for close pores in the direction of the rotated elliptical pore. By doing this, we create wrinkle connections that are flowing in the direction of the stretch. The offset is scaled depending on the amount of stretch and compression at the current surface point and we add separate user-defined constants for pores and wrinkles to help increase or decrease the tension effect. The difference in appearance when adding the tension awareness can be seen in Figure 2.

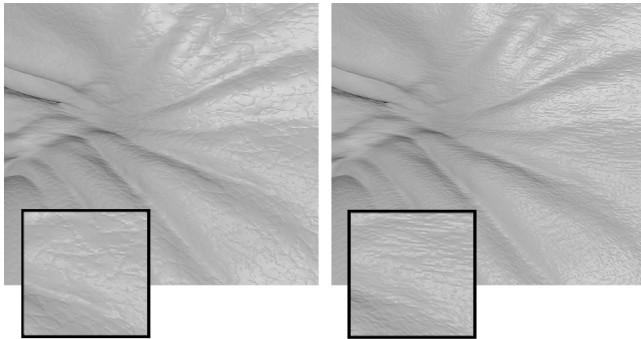


Figure 2: The appearance difference between a) skin generation without tension awareness and b) skin generation with tension awareness.

2.3 Visualisation and User Input

A valuable tool when generating skin textures is to instantly visualise the outcome of the pores and wrinkles to get an overview of how the different settings influence the resulting appearance.

We implemented visualisation options to be able to inspect the pore placements and the wrinkle connections directly in the Houdini viewport, as can be seen in Figure 3. For the pore positions, we use barycentric interpolation to find the corresponding 3D world coordinates of the generated 2D positions. To visualise the wrinkles, we utilise the index list of connected pores and append lines between them.

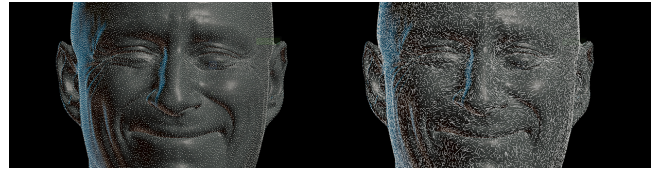


Figure 3: Showcase of the visualisation mode for pores (to the left) and for wrinkles (to the right).

2.4 Displacement Map

Once all pores and wrinkle connections have been created we sample an output displacement map. For the pores, we once again utilise a k-d tree and query for the closest pore and calculate the displacement based on the distance to its center point and the specified falloff type. The falloff function can be manipulated to create sharper or softer pore effects. To determine the displacement given by the wrinkles, we create a spatial hash and add all wrinkle lines to it. We then use the spatial hash to determine if any wrinkle lines occupy the current pixel in the output displacement map and if so, calculate their influence on the overall displacement.

3 CONCLUSIONS AND FUTURE WORK

Our plugin offers artists a wide range of settings allowing them to successfully create different skin appearances. We have evaluated our tool on subjects of various skin colors, but future work could focus on in-depth evaluations and modifications of the settings to accurately adapt to the differences across skin colors and skin types. Another aspect to explore is to create layering of wrinkles for more complex wrinkle behaviors, and to let the pore and wrinkle depth be driven by the tension of the mesh.

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